



QUALITY SYSTEM

FORM

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Revision #: 11

TITLE:

LABORATORY OOS INVESTIGATION FORM

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Phase I

Section A – Test Details

Initiator:		Test Date:		Date Initiated:	
Date of Incident:		Test Name:	Celsis Sterility Test	Sample ID:	
Name of Analyst who Performed the Test:		Sample / Active Name		Lot #:	
		Dosage Form			
Description of Incident:			☒ Client Care notified of OOS ☒ Sterility and Endotoxin test related OOS; Client must be notified within 24 hours of OOS		
	SOP / Test Method # 2.600.059	Effective Date: 03 Jan 2025	SOP / Test Method Rev: 01	Limits/Specification: <1 event	
Manager notified	Manager/designee name: Kathan Parikh	Date:	Quality Assurance Department notified:	Yes ☒ No ☐	

Section B – Investigation and Root Cause Analysis

Personnel Discrepancy		Comments
Analyst interviewed?	☒ Yes ☐ No ☐ N/A	
Correct samples analyzed?	☒ Yes ☐ No ☐ N/A	
Correct SOP / Test Method used/followed?	☒ Yes ☐ No ☐ N/A	Yes, as per SOP 2.600.059
Correct sample collection/technique?	☒ Yes ☐ No ☐ N/A	Yes, as per SOP 2.600.059
Are the Test Result Records Attached & Calculations Verified and Correct?	☒ Yes ☐ No ☐ N/A	
Is the analyst qualified to perform the test?	☒ Yes ☐ No ☐ N/A	Yes, all analysts are trained and qualified by the quality department to perform the test.
Is formula worksheet provided and verified?	☐ Yes ☐ No ☒ N/A	Not Applicable
Does the processing method/method set conform to the required specification?	☐ Yes ☐ No ☒ N/A	Not Applicable
Was the sample stored appropriately?	☒ Yes ☐ No ☐ N/A	
Material Discrepancy		Lot Number
Is the integrity of the sample(s) questionable?	☐ Yes ☒ No ☐ N/A	
Was the sample transported appropriately?	☒ Yes ☐ No ☐ N/A	
Are the other test results from this run under investigation?	☐ Yes ☒ No ☐ N/A	



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Appropriate/correct glassware used?	☐ Yes ☐ No ☒ N/A		
Are reagents stored in proper condition and not expired?	☒ Yes ☐ No ☐ N/A	Celsis Process Consumables: Information is available in EagleTrax Sample Consumables section Celsis Aliquoting Consumables: See attached Celsis data packet Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs	Celsis Process Consumables: Information is available in EagleTrax Sample Consumables section Celsis Aliquoting Consumables: See attached Celsis data packet Environmental Plates: TSA and Surface Plate: see attached environmental logs Monthly Cleaning: H2O2 strips and IPA: See attached monthly cleaning logs
Standard Used	Celsis ATP Positive Control		
Solvent Used	Not Applicable	Not Applicable	Not Applicable
Is the standard the source of the root cause?	☐ Yes ☐ No ☒ N/A		
Has the analyst been instructed to keep the original test solution for possible retesting?	☐ Yes ☐ No ☒ N/A		
Equipment Discrepancy		Equipment ID	Calibration Due Date
Was the instrument set up correctly?	☒ Yes ☐ No ☐ N/A		Mar 2026
Is the clean room facility certified?	☒ Yes ☐ No ☐ N/A		Jun 2026 Dec 2026 Jun 2026
System Performance Check (if applicable)?	☒ Yes ☐ No ☐ N/A		Mar 2026
System Suitability RSD < 2% (PDA)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
System Suitability RSD < 10% (QDA, CAD)?	☐ Yes ☐ No ☒ N/A	Not Applicable	Not Applicable
Do the environmental monitoring results conform to required standards?	☒ Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Do the controls (i.e., positive, negative, etc.) conform to the required standards?	☒ Yes ☐ No ☐ N/A	See Phase I Summary	See Phase I Summary
Is equipment/instrument functional and calibrated/qualified/certified/PM?	☒ Yes ☐ No ☐ N/A	Incubator E001034 (Sensor E001501) Incubator E001031 (Sensor E001505) Incubator E001356 (Sensor E001450) Incubator E001357 (Sensor E001449)	Jun 2026 Feb 2027 Jun 2026 Feb 2027 Jun 2026 Feb 2027 Jun 2026 Feb 2027

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Are chromatogram/spectra/raw data correct?

☐ Yes☐ No☒ N/A

Not Applicable

Not Applicable

Section C – Evaluation of Laboratory Testing/Hypothesis Testing (if applicable)☒ N/A☐ Re-injection of sample vial☐ Re-injection of original standard vial☐ Re-vialing of original working standard solution☐ Re-vialing of original working sample solution☐ Re-dilution of a working sample solution from the original stock☐ Re-dilution of a working standard solution from the original stock standard☐ Other: N/A QYC**Section D – Phase I Summary****Phase I Summary:**

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Phase I Summary (continued):

On 01 Sep 2026, the sample vials for ETX-260828-0527 were received from the Sample Submissions team and brought into the Sterile Microbiology lab. Upon arrival, each vial was sprayed with an acidified bleach disinfectant, placed into pre-disinfected bins, and allowed a 10-minute contact time. Secondary disinfection was performed in the ISO 8 anteroom (114), where the vials were again treated with acidified bleach with a 10-minute contact time before transfer into cleanroom 114B. Inside 114B, processing analyst Gabrielle Surber performed a final disinfection step with a 10-minute contact time. Once fully disinfected, the vials were transferred into the ISO 5 BSC E001316, placed on the disinfected work surface, aseptically opened, and tested in accordance with MICRO-SOP-44. Direct inoculation was performed by adding 25 mL of sample per media container, using five 300 mL FTM jars and five 300 mL TSB jars. In accordance with suitability method record 2601120497, one gram of Bovine Serum Albumin was prepared in 100 mL of PBS, and 5 mL of this solution was added to each jar. Following testing, the media bottles were transferred into incubators E001356 and E001357 to initiate incubation.

Upon completion of incubation on 08 Sep 2026, the media bottles for ETX-260828-0527 were disinfected and transferred to the middle ISO 7 buffer room (114A) for the aliquoting step per MICRO-SOP-44 (Celsis Sterility Testing). In cleanroom 114A, the media bottles were disinfected one more time before transfer into the ISO 5 BSC E001798 located in cleanroom 114A. In ISO 5 BSC E001798, the sample was aliquoted into assay cuvettes by analyst Cuong Du. Following aliquoting, Celsis Sterility Reading was performed in accordance with MICRO-SOP-44 by analyst America Alanis.

Following the reading, sample ETX-260828-0527 was found to yield a positive reading in one of the 300 mL FTM media jars (ETX-260828-0527-3/5). The average Relative Luminescence Units (RLU) from the duplicate reading tube, originating from the FTM sample bottle, yielded 7,190 RLU, which exceeded the method cutoff of 2,955 RLU, where the FTM negative control was 985 RLU. The corresponding TSB sample container ETX-260828-0527-4/5 tested negative with 1,500 RLU, below the TSB cutoff of 2,007 RLU and negative control of 669 RLU. All other FTM and TSB sample bottles in the same batch tested negative. The %CV from the duplicate reading tubes for the positive FTM bottles was well within the specification (< 30%). Additionally, all Daily Controls, including the Instrument Blank, Reagent Blank, and ATP Positive Control, were within the defined specifications, each with a %CV below 30%.

Following the OOS result, the positive FTM bottle for ETX-260828-0527 was submitted for Differential Staining and Microbial Identification under ETX-260908-0593. No microbial growth was recovered by subculture, with zero CFU observed on the plate as shown in Table 1, and no cellular morphology was visualized by differential staining. Therefore, no organism was recovered for identification or further microbiological comparison. The lack of recovery on subculture does not invalidate the initial analytical Celsis result and may be consistent with low-level or stressed microbial cells under the test conditions, though a specific physiological state could not be confirmed.

The culture media utilized were within their expiry period. The negative culture media bottles for the direct inoculation method for the original culture were handled, processed, and incubated in a manner identical to that of actual samples. No microbial growth was observed in the corresponding negative control.

Monthly cleaning and disinfection of the outermost ISO 8 Anteroom, the middle ISO 7 Buffer room, the innermost ISO 7 cleanroom, and its containing ISO 5 Biosafety Cabinets for CR114 was performed on 30 Aug 2026, as per MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology). It was documented that all H2O2 indicators passed. This confirms the efficient monthly cleaning of all three parts of Cleanroom 114.

Attached Tables 2 and 3 present the environmental monitoring results for the duration of testing. The environmental monitoring (EM) plates were incubated for no less than 48 hours at 30-35°C and no less than an additional five days at 20-25°C as per MICRO-SOP-2 (Environmental Monitoring of the Cleanroom Facility). Table 2 pertains to Environmental Monitoring performed during Celsis Sterility Processing, and Table 3 pertains to Environmental Monitoring performed during Celsis Sterility Aliquoting.

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Phase I Summary (continued):

A review of the Environmental Monitoring (EM) results for the relevant Celsis Sterility processing period showed no microbial growth on personnel monitoring plates, settling plates, or ISO 5 BSC E001316 surface sampling plates on the date of testing (01Sep2026), the preceding sampling date (31Aug2026), or the subsequent sampling date (02Sep2026). Weekly active air and surface sampling of the cleanroom suite associated with processing was performed on 04Sep2026. No microbial recovery was observed from weekly surface sampling of the cleanroom suite. Active air sampling showed recovery of 1 CFU (ETX-260914-0487) in the ISO 8 area, identified as Gram-positive coccobacilli and Gram-positive rods, with zero recovery observed in the ISO 7 buffer or cleanrooms.

Similarly, after reviewing the Environmental Monitoring (EM) results for the relevant Celsis Sterility aliquoting period, no microbial growth was observed on personnel monitoring plates, ISO 5 BSC E001798 surface sampling plates, or settling plates on the date of aliquoting (08Sep2026), the preceding sampling date (07Sep2026), or the subsequent sampling date (09Sep2026). Weekly active air sampling of the cleanroom suite performed on 10Sep2026 showed recovery of 1 CFU (ETX-260921-0520) in the ISO 8 area, identified as Gram-positive cocci, with zero recovery in the ISO 7 buffer or cleanrooms, and weekly surface sampling of the cleanroom suite showed no growth.

Evaluation of Environmental Monitoring Results: It is important to note that, for both processing and aliquoting activities, the samples are processed within validated ISO 5 BSCs (E001316 and E001798 respectively). Although microbial recovery was observed from weekly active air monitoring in the outermost ISO 8 room 114 background environment in cleanroom suite (CR114), these recoveries were not obtained from the ISO 5 processing environment or the ISO 7 cleanroom environment.

The absence of microbial recovery from both the processing and the aliquoting analyst glove plates, ISO 5 BSC surface monitoring plates, and ISO 5 settling plates indicates that no identifiable contamination transfer pathway existed between the monitored background cleanroom environment and the ISO 5 critical processing areas. Samples were also prepared and transferred into disinfected, lidded containers and transported on carts within the controlled cleanroom environment. Therefore, while the active air results demonstrate microbial recovery in the surrounding ISO 8 area, there is no evidence that the recovery originated from, or affected, the ISO 5 BSC environment where the test sample was handled and processed.

All analysts confirmed full compliance with cleaning procedures as outlined in MICRO-SOP-9 (Cleaning and Disinfecting Procedure for Microbiology) and MICRO-SOP-44 (Celsis Sterility Testing). A review of the available data confirms that the cleanroom and equipment conditions remained within acceptable parameters. No deviations or obvious signs of laboratory error during processing and aliquoting were noted from analysts.

No other sample processed by analyst Gabrielle Surber on 01 Sep 2026 and aliquoted by analyst Cuong Du on 08 Sep 2026, failed Celsis Sterility testing that day. To assess the potential for sample-to-sample contamination contributing to the positive results, a comprehensive review was conducted of all samples processed in the same batch. All other samples processed by the same analyst on the day of testing were found to test negative for microbial growth. These findings suggest that cross-contamination between samples is unlikely.

An analysis of the six-month sample history for Optimal Balance Pharmacy indicates that Eagle Analytical processed [X] samples for Celsis Sterility testing with no prior occurrences of an out of specification or positive result during this period.

A review of the lot history shows that there have been no additional submissions of sample lot LG342010349 for retesting using Celsis Sterility Testing or any other sterility method.

Based on the laboratory investigation, no assignable laboratory cause was identified involving the analyst, instrument, reagents, supplies, or monitored laboratory environment. The initial Celsis OOS result therefore remains valid in accordance with the applicable laboratory OOS procedure.



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Has root cause been identified as a possible laboratory error?*Note: Root cause must be conclusive, verifiable, and well-documented.*☐ Yes☐ If retesting is **required**, complete Phase II of the OOS (initiate Form 3.100.019.F02).☒ No, the OOS result is valid.**Reviewer comment(s):****Prepared by (Print):**

Qiyue Chen

Signature/Date:

Lab Manager/Designee (Print):

Signature/Date:

Section E – Quality Disposition

Can investigation be closed?☐ Yes☐ Original result is valid.☐ No, initiate Form 3.100.019.F02.**Approved by QA Representative (Print):**

Signature/Date: